

Saeid Samadi

SENIOR AUTONOMY ENGINEER | REAL-TIME SYSTEMS | CONTROL & SENSING

United Kingdom sa.samadi74@gmail.com saeidsamadi.github.io [LinkedIn](#) [GitHub](#) [Google Scholar](#)

PROFILE

Senior autonomy engineer with 9+ years across humanoid control, contact-rich manipulation, teleoperation, force/torque sensing, and real-robot validation. At Humanoid, owns control capabilities from whole-body control and impedance tuning through embodiment integration, data collection, and industrial task deployment. Ph.D. in multi-contact humanoid control, with a track record of building the robot-side systems, data loops, and cross-functional engineering required to turn learned policies into reliable physical behaviour.

CORE EXPERTISE

Manipulation & touch	Contact-rich and bimanual manipulation; teleoperation; force/torque and capacitive sensing; gripper and suction integration; robot calibration; real-robot data collection.
Control & planning	Whole-body and optimisation-based control; impedance, compliant and force control; kinematics and dynamics; balance/CoM constraints; collision avoidance; motion and footstep planning.
Learning & systems	VLA/RL policy integration; sim-to-real validation; episode and log analysis; parameter identification; robot bring-up; safety, diagnostics, and cross-layer root-cause analysis.
Software & tools	C++, Python, ROS/ROS2, Linux, Git, CMake, Docker, CI; MuJoCo, Gazebo, Choreonoid, mc_rtc, MoveIt, RViz, MATLAB/Simulink.
Robot platforms	Full-scale humanoid platforms, TALOS, HRP-4, and dual-arm mobile manipulators; force/torque sensors, grippers, suction end-effectors, and embedded interfaces.

PROFESSIONAL EXPERIENCE

Senior Autonomy Engineer Humanoid	May 2026–Present United Kingdom
- Own whole-body and manipulation-control capabilities for full-scale humanoid robots, taking work from algorithm and architecture through C++/ROS2 implementation, simulation, code review, hardware validation, and operational rollout. - Developed task-specific control behaviours for contact-rich parcel handling, including alignment, bimanual box handling, suction-gripper integration, safe mode transitions, and maintenance of task constraints; changes increased execution success and data-collection reliability. - Led cross-layer root-cause analysis of teleoperation jumps and missed command cycles by correlating tracking, network delivery, onboard scheduling, ROS/DDS timing, server output, and WBC logs; isolated upstream command gaps and helped drive smoothness fixes into the main stack. - Diagnosed and resolved whole-body-control performance bottlenecks, and am implementing a safety-conscious, dynamics-aware WBC architecture for high-frequency real-time deployment on hardware. - Validated mechanical range of motion and translated findings into robot limits and safe controller constraints; coordinated with hardware and learned-policy teams to expand usable manipulation workspace while preserving policy compatibility. - Tuned and integrated impedance, stiffness, damping, gravity/torque feedforward, and calibration changes to improve arm tracking and manipulation authority across robot configurations.	
Robotics Laboratory Manager University of Edinburgh	Jun 2023–May 2026 Edinburgh, UK
- Led operational and technical delivery across eight robotics laboratories supporting humanoid, mobile-manipulation, assistive, and learning-based robotics programmes. - Owned hardware/software readiness, safety processes, procurement, vendor coordination, equipment lifecycle, experimental infrastructure, and technical support for demonstrations, workshops, student projects, industry visits, and external collaborations. - Built repeatable support and maintenance processes and received the 2025 School of Informatics Staff Award for Outstanding Contribution.	

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PROFESSIONAL EXPERIENCE CONTINUED

Postdoctoral Researcher

University of Edinburgh · SLMC Group

Mar 2022–Jun 2023

Edinburgh, UK

- Developed control and navigation components for assistive robots operating in dynamic healthcare and human-centred environments within an EU Horizon project.
- Contributed to TALOS humanoid control, whole-body motion generation, online multi-contact planning, force-aware sensor integration, and compliant dual-arm mobile manipulation.
- Converted research concepts into robust software prototypes, experiments, publications, and live demonstrations with interdisciplinary partners.

Doctoral Researcher in Humanoid Control

LIRMM · University of Montpellier

Oct 2018–Feb 2022

Montpellier, France

- Owned the end-to-end algorithm-to-robot pipeline for real-time whole-body balance control on HRP-4 under fixed, sliding, and mixed unilateral contacts.
- Designed control formulations, simulation studies, force/torque sensor integration, experiment plans, and real-hardware validation for contact-aware multi-contact stability.
- Published first-author work in IEEE Robotics and Automation Letters and ICRA and completed a Ph.D. thesis on humanoid balance in multi-contact mode settings.

SELECTED SYSTEMS IMPACT

- **Contact-rich robot intelligence:** built interfaces and experiments spanning F/T sensing, compliant control, teleoperation, grippers and suction tooling, learned-policy execution, and episode-level diagnostics—the physical foundation required to collect and evaluate high-quality visuo-tactile manipulation data.
- **Whole-body manipulation platform:** integrated balance, collision avoidance, CoM and manipulability constraints, joint and actuator limits, and Cartesian task commands into deployable humanoid behaviour.
- **Experiment and data quality:** designed test matrices and data-collection workflows across robot configurations, compared episodes and logs across stacks, and turned failure signatures into actionable changes for control, software, and policy teams.
- **Cross-functional ownership:** operated across algorithms, production software, hardware bring-up, lab infrastructure, safety, procurement, vendors, data collection, demonstrations, and technical handover.

EDUCATION

Ph.D. in Robotics and Artificial Intelligence

University of Montpellier · Thesis: Humanoid Robots Balance in Multi-Contact Modes Settings

2018–2022

M.Sc. in Mechanical Engineering

University of Tehran · Thesis: Online Walking of a Humanoid Robot on Rough Surfaces with Different Heights

2016–2018

B.Sc. in Mechanical Engineering

University of Tabriz

2012–2016

SELECTED PUBLICATIONS

Humanoid Control Under Interchangeable Fixed and Sliding Unilateral Contacts

S. Samadi, J. Roux, A. Tanguy, S. Caron, A. Kheddar · IEEE Robotics and Automation Letters, 2021

Online Multi-Contact Receding Horizon Planning via Value Function Approximation

J. Wang, S. Kim, T. S. Lembono, W. Du, J. Shim, S. Samadi, et al. · IEEE Transactions on Robotics, 2024

NAS: N-Step Computation of All Solutions to the Footstep Planning Problem

S. Samadi, J. Wang, H. Wang, et al. · IEEE-RAS International Conference on Humanoid Robots, 2024

Full list: Google Scholar

AWARDS, SERVICE & LANGUAGES

- Staff Award for Outstanding Contribution, University of Edinburgh School of Informatics, 2025.
- UK Global Talent Visa, UKRI route, awarded for contributions to an EU Horizon-funded robotics project, 2022.
- Reviewer for IEEE T-RO, RA-L, ICRA, IROS, and Humanoids, 2018–Present.

Languages: Professional English; fluent Persian, Azeri, and Turkish; intermediate French and Arabic; basic German.